Python for Data Science / Data Analytics

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85+ Multiple Choice Questions with Answers

Q1. What is the primary data structure in Pandas for 2D data?

- A) List
- B) Array
- C) DataFrame
- D) Series

Answer: C

Q2. How do you select a single column 'Age' from a DataFrame df?

- A) df['Age'] only
- B) df.Age only
- C) Both A and B
- D) df.select('Age')

Answer: C

Q3. Which method is used to read a CSV file in Pandas?

- A) pd.load_csv()
- B) pd.read_csv()
- C) pd.import_csv()
- D) pd.open_csv()

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Answer: B

Q4. What does df.head() return by default?

- A) First 10 rows
- B) First 5 rows
- C) First row
- D) All rows

Answer: B

Q5. Which method returns the number of rows and columns in a DataFrame?

- A) df.dimensions()
- B) df.size
- C) df.shape
- D) df.count()

Answer: C

Q6. How do you rename a column 'old_name' to 'new_name' in a DataFrame?

- A) df.rename(columns={'old_name': 'new_name'})
- B) df.change_column('old_name', 'new_name')

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- C) df['old_name'].name = 'new_name'
- D) df.columns.rename('old_name', 'new_name')

Answer: A

Q7. What is the difference between loc and iloc in Pandas?

- A) No difference
- B) loc uses labels, iloc uses integer positions
- C) loc is faster than iloc
- D) iloc is deprecated

Answer: B

Q8. Which method is used to sort a DataFrame by a column 'Age'?

- A) df.sort('Age')
- B) df.order_by('Age')
- C) df.sort_values('Age')
- D) df.arrange('Age')

Answer: C

Q9. How do you get unique values in a column 'City'?

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- A) df['City'].unique()
- B) df['City'].distinct()
- C) df['City'].get_unique()
- D) unique(df['City'])

Answer: A

Q10. What does df.describe() return?

- A) First few rows
- B) Data types of columns
- C) Statistical summary of numerical columns
- D) Column names

Answer: C

Q11. What is the default data type of a NumPy array created from integers?

- A) float64
- B) int32 or int64 (platform dependent)
- C) object
- D) string

Answer: B

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Q12. How do you create a 1D NumPy array with values from 0 to 9?

- A) np.array(range(10))
- B) np.arange(10)
- C) Both A and B
- D) np.range(10)

Answer: C

Q13. What is the shape of np.zeros((3, 4))?

- A) (3, 4)
- B) (4, 3)
- C) 12
- D) (3,)

Answer: A

Q14. Which operation is element-wise in NumPy?

- A) arr1 + arr2
- B) arr1 * arr2
- C) arr1 / arr2

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- D) All of the above

Answer: D

Q15. How do you reshape a 1D array of 12 elements to a 3x4 matrix?

- A) arr.reshape(3, 4)
- B) arr.shape(3, 4)
- C) arr.resize(3, 4)
- D) arr.transform(3, 4)

Answer: A

Q16. What does np.linspace(0, 10, 5) create?

- A) [0, 2, 4, 6, 8]
- B) [0, 2.5, 5, 7.5, 10]
- C) [0, 1, 2, 3, 4]
- D) [0, 10, 20, 30, 40]

Answer: B

Q17. Which method is used to find the maximum value in a NumPy array?

- A) arr.max()

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- B) np.max(arr)
- C) Both A and B
- D) arr.maximum()

Answer: C

Q18. What is broadcasting in NumPy?

- A) Sending arrays over network
- B) Automatic expansion of arrays for element-wise operations
- C) Converting arrays to lists
- D) Parallel processing

Answer: B

Q19. Which method removes duplicate rows from a DataFrame?

- A) df.remove_duplicates()
- B) df.drop_duplicates()
- C) df.unique()
- D) df.deduplicate()

Answer: B

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Q20. How do you check for missing values in a DataFrame?

- A) df.isnull()
- B) df.isna()
- C) Both A and B
- D) df.missing()

Answer: C

Q21. What does df.dropna() do by default?

- A) Drops rows with any missing values
- B) Drops columns with any missing values
- C) Fills missing values with 0
- D) Replaces missing values with mean

Answer: A

Q22. Which parameter in dropna() specifies the minimum number of non-NA values required?

- A) minimum
- B) thresh
- C) limit

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- D) count

Answer: B

Q23. How do you replace all occurrences of 'Unknown' with NaN in a DataFrame?

- A) df.replace('Unknown', np.nan)
- B) df.substitute('Unknown', np.nan)
- C) df.change('Unknown', np.nan)
- D) df.convert('Unknown', np.nan)

Answer: A

Q24. What is the purpose of df.astype() method?

- A) To sort data
- B) To convert data types of columns
- C) To filter data
- D) To merge dataframes

Answer: B

Q25. How do you remove leading and trailing whitespaces from string columns?

- A) df['column'].strip()

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- B) df['column'].str.strip()
- C) df.strip('column')
- D) strip(df['column'])

Answer: B

Q26. Which method fills missing values with a specified value?

- A) df.fill()
- B) df.fillna()
- C) df.replace_na()
- D) df.impute()

Answer: B

Q27. How do you fill missing values with the mean of the column?

- A) df.fillna(df.mean())
- B) df.fillna(mean())
- C) df.fill_mean()
- D) df.impute_mean()

Answer: A

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Q28. What does the method parameter in fillna() do?

- A) Specifies the filling value
- B) Specifies forward fill or backward fill
- C) Specifies the column to fill
- D) Specifies the data type

Answer: B

Q29. Which value does 'ffill' represent in fillna(method='ffill')?

- A) Fill with first value
- B) Forward fill (propagate last valid observation forward)
- C) Fill with mean
- D) Fill with zero

Answer: B

Q30. How do you interpolate missing values in a time series?

- A) df.interpolate()
- B) df.fillna(interpolate=True)
- C) df.impute()

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- D) df.smooth()

Answer: A

Q31. What does df.dropna(axis=1) do?

- A) Drops rows with missing values
- B) Drops columns with missing values
- C) Fills missing values
- D) Counts missing values

Answer: B

Q32. How do you fill missing values differently for each column?

- A) Pass a dictionary to fillna()
- B) Use a loop
- C) Both A and B
- D) Not possible

Answer: C

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Q33. What is the purpose of feature scaling?

- A) To remove outliers
- B) To bring features to similar scales
- C) To reduce dimensionality
- D) To encode categorical variables

Answer: B

Q34. Which scaling method transforms features to a range between 0 and 1?

- A) StandardScaler
- B) MinMaxScaler
- C) RobustScaler
- D) Normalizer

Answer: B

Q35. What does StandardScaler do?

- A) Scales to [0, 1] range
- B) Scales to have mean=0 and std=1
- C) Scales to [-1, 1] range
- D) Removes outliers

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Answer: B

Q36. Which scaler is most robust to outliers?

- A) MinMaxScaler
- B) StandardScaler
- C) RobustScaler
- D) MaxAbsScaler

Answer: C

Q37. From which library do you import StandardScaler in scikit-learn?

- A) sklearn.preprocessing
- B) sklearn.scaling
- C) sklearn.transform
- D) sklearn.normalize

Answer: A

Q38. What is normalization in the context of feature scaling?

- A) Same as standardization
- B) Scaling individual samples to have unit norm

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- C) Removing duplicates
- D) Handling missing values

Answer: B

Q39. When should you use MinMaxScaler?

- A) When you want features in a bounded interval
- B) When data has outliers
- C) When data is normally distributed
- D) Never

Answer: A

Q40. What happens if you don't scale features before applying KNN or SVM?

- A) Nothing, works the same
- B) Features with larger scales dominate the distance calculations
- C) Algorithm won't run
- D) Improves performance

Answer: B

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Q41. What is Label Encoding?

- A) Encoding numerical values
- B) Converting categorical labels to numerical values
- C) Creating new features
- D) Removing labels

Answer: B

Q42. Which sklearn class is used for Label Encoding?

- A) LabelEncoder
- B) OneHotEncoder
- C) OrdinalEncoder
- D) CategoricalEncoder

Answer: A

Q43. What is the main disadvantage of Label Encoding for nominal categorical variables?

- A) Slow performance
- B) Creates ordinal relationship that doesn't exist
- C) Cannot handle missing values
- D) Uses too much memory

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Answer: B

Q44. What does One-Hot Encoding do?

- A) Assigns single number to each category
- B) Creates binary columns for each category
- C) Removes categorical variables
- D) Sorts categories

Answer: B

Q45. If a categorical feature has 5 unique values, how many columns will One-Hot Encoding create?

- A) 4
- B) 5
- C) 6
- D) 1

Answer: B (Note: With drop_first=True, it would create 4 columns)

Q46. Which pandas method is commonly used for One-Hot Encoding?

- A) pd.encode()

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- B) pd.get_dummies()
- C) pd.one_hot()
- D) pd.categorize()

Answer: B

Q47. What is the "dummy variable trap"?

- A) Too many categories
- B) Perfect multicollinearity from including all dummy variables
- C) Missing values in encoding
- D) Encoding errors

Answer: B

Q48. When should you use Label Encoding over One-Hot Encoding?

- A) For nominal variables
- B) For ordinal variables with inherent order
- C) For all categorical variables
- D) Never

Answer: B

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Q49. How do you merge two DataFrames df1 and df2 on a common column 'ID'?

- A) pd.merge(df1, df2, on='ID')
- B) df1.merge(df2, on='ID')
- C) Both A and B
- D) df1.join(df2, key='ID')

Answer: C

Q50. What is the difference between merge() and concat()?

- A) No difference
- B) merge() combines on keys, concat() stacks along an axis
- C) concat() is faster
- D) merge() only works with 2 DataFrames

Answer: B

Q51. How do you group data by a column and calculate mean?

- A) df.groupby('column').mean()
- B) df.group('column').mean()

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- C) df.aggregate('column', 'mean')
- D) df.summarize('column', 'mean')

Answer: A

Q52. What does df.pivot_table() do?

- A) Rotates the DataFrame
- B) Creates a spreadsheet-style pivot table
- C) Sorts the data
- D) Filters the data

Answer: B

Q53. How do you apply a custom function to each element of a DataFrame column?

- A) df['column'].map(function)
- B) df['column'].apply(function)
- C) Both A and B
- D) df['column'].transform(function)

Answer: C

Q54. What is the purpose of the axis parameter in pandas operations?

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- A) To specify rotation
- B) To specify whether to operate row-wise (0) or column-wise (1)
- C) To specify data type
- D) To specify sorting order

Answer: B

Q55. How do you convert a pandas Series to a NumPy array?

- A) series.to_array()
- B) series.values
- C) np.array(series)
- D) Both B and C

Answer: D

Q56. What does pd.crosstab() compute?

- A) Cross-validation scores
- B) Cross-tabulation of two or more factors
- C) Correlation matrix
- D) Covariance matrix

Answer: B

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Q57. How do you save a DataFrame to a CSV file?

- A) df.save_csv('file.csv')
- B) df.to_csv('file.csv')
- C) df.write_csv('file.csv')
- D) df.export_csv('file.csv')

Answer: B

Q58. What is vectorization in pandas/numpy?

- A) Converting to vectors
- B) Performing operations on entire arrays without explicit loops
- C) 3D operations
- D) Parallel processing

Answer: B

Q59. How do you filter rows where Age > 25 in a DataFrame?

- A) df[df['Age'] > 25]
- B) df.filter(Age > 25)
- C) df.select(Age > 25)

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- D) df.where('Age' > 25)

Answer: A

Q60. What does df.corr() calculate?

- A) Covariance matrix
- B) Correlation matrix
- C) Confusion matrix
- D) Distance matrix

Answer: B

Q61. What is the output of [1, 2, 3] + [4, 5]?

- A) [1, 2, 3, 4, 5]
- B) [5, 7, 3]
- C) Error
- D) [[1, 2, 3], [4, 5]]

Answer: A

Q62. What does list.pop() do without any argument?

- A) Removes first element
- B) Removes last element and returns it

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- C) Removes all elements
- D) Returns the list

Answer: B

Q63. What is the output of [1, 2, 3, 4, 5][1:4]?

- A) [1, 2, 3, 4]
- B) [2, 3, 4]
- C) [2, 3, 4, 5]
- D) [1, 2, 3]

Answer: B

Q64. Which method adds an element to the end of a list?

- A) list.add()
- B) list.insert()
- C) list.append()
- D) list.push()

Answer: C

Q65. What is the time complexity of accessing an element by index in a Python list?

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- A) O(n)
- B) O(log n)
- C) O(1)
- D) $O(n^2)$

Answer: C

Q66. What is the output of {1, 2, 3} & {2, 3, 4}?

- A) {1, 2, 3, 4}
- B) {2, 3}
- C) {1, 4}
- D) Error

Answer: B (Intersection operation)

Q67. Which characteristic is TRUE about Python sets?

- A) Ordered collection
- B) Allows duplicate elements
- C) Mutable (can add/remove elements)
- D) Supports indexing

Answer: C

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Q68. What does set1.difference(set2) return?

- A) Elements in both sets
- B) Elements in set1 but not in set2
- C) Elements in set2 but not in set1
- D) All elements from both sets

Answer: B

Q69. Which operator performs union of two sets?

- A) &
- B) |
- C) ^
- D) -

Answer: B

Q70. What is the output of len({1, 2, 2, 3, 3, 3})?

- A) 6
- B) 3
- C) 4
- D) Error

Answer: B (Sets don't allow duplicates)

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Q71. What is the output of {'a': 1, 'b': 2}.get('c', 0)?

- A) None
- B) 0
- C) Error
- D) 'c'

Answer: B

Q72. Which method returns all keys in a dictionary?

- A) dict.keys()
- B) dict.getkeys()
- C) dict.all_keys()
- D) dict.keylist()

Answer: A

Q73. What happens when you access a non-existent key using dict['key']?

- A) Returns None
- B) Returns 0
- C) Raises KeyError

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- D) Returns empty string

Answer: C

Q74. How do you merge two dictionaries dict1 and dict2 (Python 3.9+)?

- A) dict1 + dict2
- B) dict1 | dict2
- C) dict1.merge(dict2)
- D) merge(dict1, dict2)

Answer: B

Q75. What is the time complexity of dictionary lookup by key?

- A) O(n)
- B) O(log n)
- C) O(1) average case
- D) $O(n^2)$

Answer: C

Q76. What does the following code output?

```
def func(x=[]):
```

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```
    x.append(1)
    return x
print(func())
print(func())
```

- A) [1] and [1]
- B) [1] and [1, 1]
- C) Error
- D) [] and [1]

Answer: B (Mutable default argument trap)

Q77. Which approach is most efficient to check if an element exists in a collection?

- A) Using a list with in operator
- B) Using a set with in operator
- C) Using a dictionary with in operator
- D) Both B and C

Answer: D (Both are O(1) average case vs O(n) for list)

Q78. What is the output of the following?

```
x = [1, 2, 3]
```

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```
y = x
y.append(4)
print(x)
```

- A) [1, 2, 3]
- B) [1, 2, 3, 4]
- C) [4]
- D) Error

Answer: B (y is a reference to x, not a copy)

Q79. How do you find the most frequent element in a list efficiently?

- A) Use max() with count()
- B) Use collections.Counter
- C) Sort and iterate
- D) Use a dictionary manually

Answer: B (Most Pythonic and efficient)

Q80. What does this list comprehension do: [x**2 for x in range(10) if x % 2 == 0]?

- A) Squares of all numbers from 0 to 9
- B) Squares of even numbers from 0 to 8

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- C) Even numbers from 0 to 9
- D) Squares of odd numbers

Answer: B

Q81. Which function is used to create a figure and subplots in Matplotlib?

- A) plt.figure()
- B) plt.subplots()
- C) plt.create_plot()
- D) plt.subplot()

Answer: B (plt.subplots() returns both figure and axes objects)

Q82. What is the main advantage of Seaborn over Matplotlib?

- A) Faster performance
- B) Better default aesthetics and statistical plotting capabilities
- C) More plot types
- D) Works without Matplotlib

Answer: B

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Q83. Which Seaborn function is best for visualizing the distribution of a numerical variable?

- A) sns.barplot()
- B) sns.scatterplot()
- C) sns.histplot() or sns.kdeplot()
- D) sns.lineplot()

Answer: C

Q84. How do you create a correlation heatmap in Seaborn?

- A) sns.heatmap(df.corr())
- B) sns.corrplot(df)
- C) sns.correlation(df)
- D) plt.heatmap(df)

Answer: A

Q85. Which parameter in Seaborn's pairplot() is used to color points by a categorical variable?

- A) color
- B) palette

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- C) hue
- D) category

Answer: C

Q86. What is the output of the following code?

```
import pandas as pd
df = pd.DataFrame({'A': [1, 2, 3], 'B': [4, 5, 6]})
print(df.loc[0, 'A'])
```

- A) 0
- B) 1
- C) [1, 4]
- D) Error

Answer: B (loc[0, 'A'] accesses row with index 0, column 'A', which is 1)

Q87. In NumPy, what does the axis parameter represent in np.sum(arr, axis=0)?

- A) Sum along rows (column-wise sum)
- B) Sum along columns (row-wise sum)
- C) Sum of all elements
- D) Sum of diagonal elements

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Answer: A (axis=0 means operation along the first dimension, collapsing rows, resulting in column-wise sum)

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